Machine Learning Midterm Project

Course: Introduction to Machine Learning

1 Project Overview

The purpose of this project is to give students practical experience with the complete machine learning workflow.

Students will design and implement a full pipeline including:

- exploratory data analysis
- data preprocessing
- model training
- performance evaluation
- error analysis

This project focuses on **classical machine learning methods**.

2 Team Policy

Students must work in teams of **two students**.

All team members are expected to contribute equally to the project.

3 Dataset

The dataset contains records of students in a hypothetical country. Each record describes a student's academic and socio-economic attributes.

The goal is to predict whether a student receives a scholarship.

4 Project Tasks

4.1 Exploratory Data Analysis

Students should:

- examine the dataset structure
- compute descriptive statistics
- visualize important variables
- identify potential data quality issues

4.2 Data Preprocessing

Typical preprocessing steps may include:

- handling missing values
- encoding categorical variables
- feature normalization or scaling

Students should justify all preprocessing decisions.

4.3 Model Training

Students must train at least **two machine learning models**.
Examples include:

- logistic regression
- decision trees
- random forest
- support vector machines

4.4 Model Evaluation

Students should evaluate models using appropriate metrics.

Typical metrics include:

- accuracy
- precision
- recall
- F1 score

4.5 Error Analysis

Students should analyze prediction errors and discuss possible reasons.

5 Submission Requirements

Each team must submit the following files:

- `report.pdf`
- `notebook.ipynb`
- `predictions.csv`

6 Hidden Test Evaluation

Model performance will be evaluated on a hidden test dataset. Students will receive the test inputs but not the labels.

The prediction file must follow the format described in the `submission_rules` document.